



AgriML

AI-Powered Fertilizer Recommendation System

Machine Learning • Climate-Aware • Sustainable Farming

Client-Side • Browser-Based • Zero Server Dependencies

The Problem

Why do farmers need intelligent fertilizer recommendations?

Key Challenges

- Over-fertilization wastes money and pollutes groundwater
- Under-fertilization leads to poor crop yields and food insecurity
- Nutrient imbalance — excess of one nutrient blocks others
- Climate variability changes optimal timing & quantity
- Manual soil testing is expensive for small-hold farmers
- No easy agronomist access in rural India

30%

of global fertilizer wasted

₹8,000 Cr

annual loss to Indian farmers

10 Crops

covered by AgriML

7 Types

fertilizer classes

Our Solution: AgriML

An end-to-end ML pipeline running entirely in the browser



Data Input

Soil NPK, pH, moisture, temperature, humidity, crop type



ML Engine

Random Forest Classifier + Gradient Boosting Regressor



Optimization

Quantity, cost & environmental impact scoring



Climate Aware

7-day forecast adjusts timing and quantity in real-time

System Architecture

12 interconnected modules — 100% client-side JavaScript

Data Layer

- dataset.js — 2,000 synthetic samples
- preprocessing.js — imputation + normalization
- fertilizerModel.js — Random Forest
- yieldModel.js — Gradient Boosting

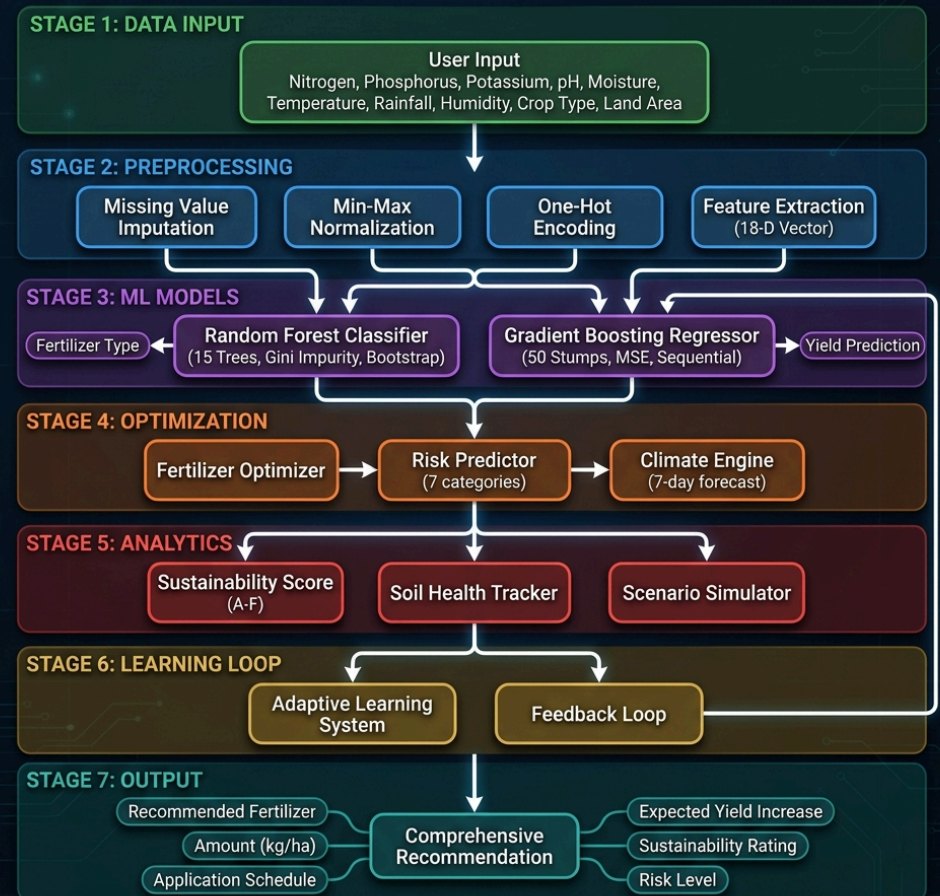
Analysis Layer

- climateEngine.js — 7-day forecast
- optimizer.js — qty, cost & env
- riskPredictor.js — 7 risk categories
- sustainability.js — 6-component index

Learning Layer

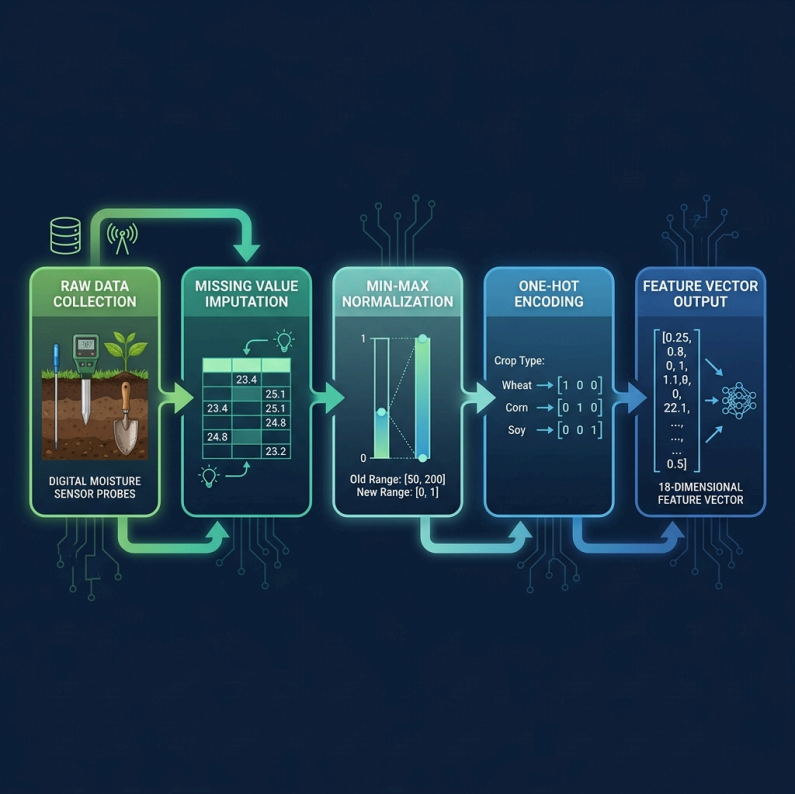
- adaptiveLearning.js — correction factors
- feedbackLoop.js — farmer feedback
- soilHealth.js — degradation tracker

AgriML - AI Model Architecture Flowchart



Data Pipeline & Preprocessing

From raw sensor data to 18-dimensional model-ready feature vector



1

Synthetic Data

2,000 samples 10 crops × 7 fertilizers

2

Imputation

Mean/Mode fill missing values

3

Normalization

Min-Max scaling to [0,1] range

4

One-Hot Encode

10 binary columns for crop type

5

Feature Vector

18-D combined vector → model

Input Features

8 numeric features + 1 categorical feature (crop type)

Feature	Type	Unit	Range	Description
Nitrogen (N)	Numeric	kg/ha	0–200	Available nitrogen in soil
Phosphorus (P)	Numeric	kg/ha	0–150	Available phosphorus in soil
Potassium (K)	Numeric	kg/ha	0–120	Available potassium in soil
pH	Numeric	—	3.0–10.0	Soil acidity / alkalinity
Moisture	Numeric	%	5–100	Soil moisture percentage
Temperature	Numeric	°C	-5–50	Ambient temperature
Rainfall	Numeric	mm	0–450	Monthly rainfall
Humidity	Numeric	%	10–100	Air humidity percent
Crop Type	Categorical	—	10 types	Rice, Wheat, Maize, Cotton, ...

Supported Crop Profiles

Optimal NPK, pH, temperature and yield ranges for each crop

Crop	N (kg/ha)	P (kg/ha)	K (kg/ha)	pH	Temp (°C)	Yield (t/ha)
Rice	60-120	30-60	30-60	5.5-7.0	22-35	3.0-8.0
Wheat	80-140	40-70	20-50	6.0-7.5	12-25	2.5-6.5
Maize	80-160	30-60	20-50	5.8-7.0	18-32	3.0-9.0
Cotton	60-120	20-50	20-40	6.0-7.5	20-35	1.5-4.0
Sugarcane	100-200	40-80	40-80	5.5-7.5	22-36	50-120
Soybean	20-50	40-70	20-50	6.0-7.0	20-30	1.5-4.0
Potato	80-150	50-90	60-100	5.0-6.5	15-25	15-40
Tomato	60-130	40-80	50-90	5.5-7.0	18-30	20-60
Groundnut	10-30	30-60	20-40	5.5-7.0	22-33	1.0-3.5
Barley	60-110	30-55	20-45	6.0-8.0	10-22	2.0-5.5

Model 1: Random Forest Classifier

Fertilizer type recommendation — ensemble of 15 decision trees

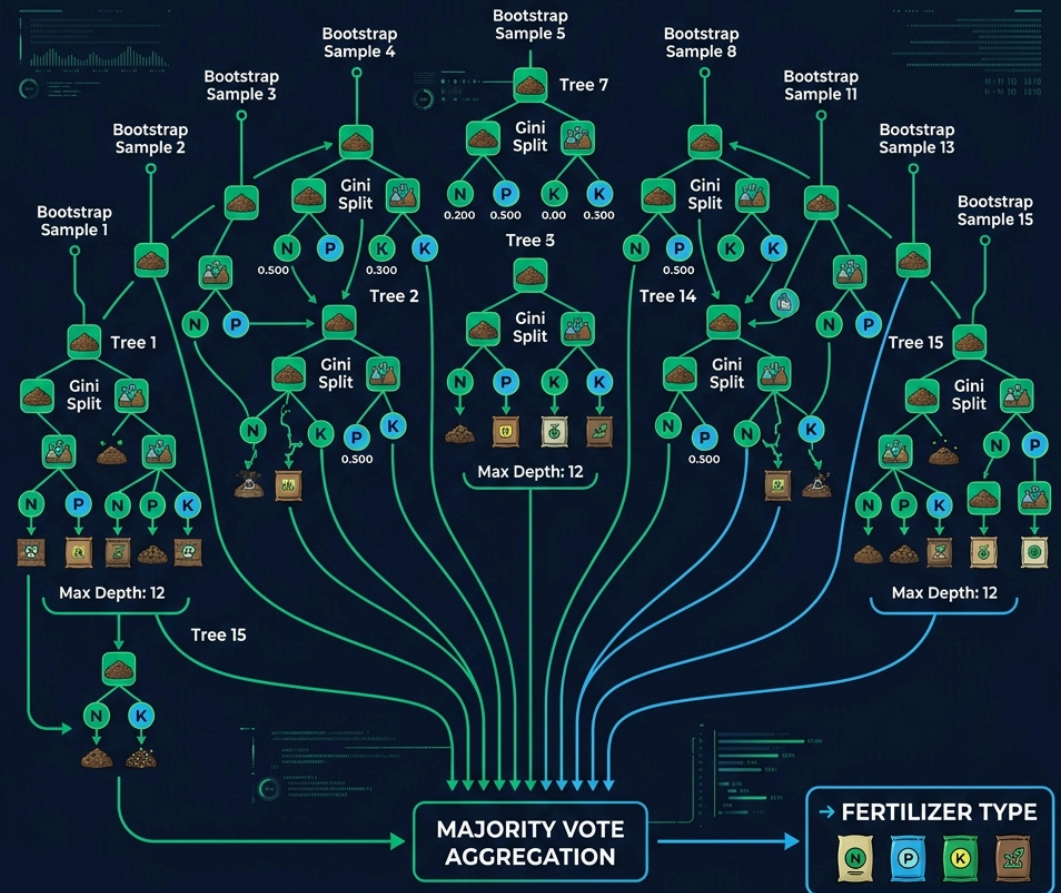
How It Works

- Bootstrap Sampling — each tree gets a random sample (with replacement)
- Random Feature Subset — 60% of features per split for diversity
- Gini Impurity — measures split quality at each node
- Majority Vote — aggregates predictions from all 15 trees
- Confidence Score — % of trees agreeing on the top class
- 7 Output Classes — one per fertilizer type

Key Params

Parameter	Value
Trees	15
Max Depth	12
Min Samples/Leaf	3
Feature Ratio	60%
Criterion	Gini

RANDOM FOREST ENSEMBLE MODEL: FERTILIZER TYPE PREDICTION



Model 2: Gradient Boosting Regressor

Yield prediction using 50 sequential decision stumps

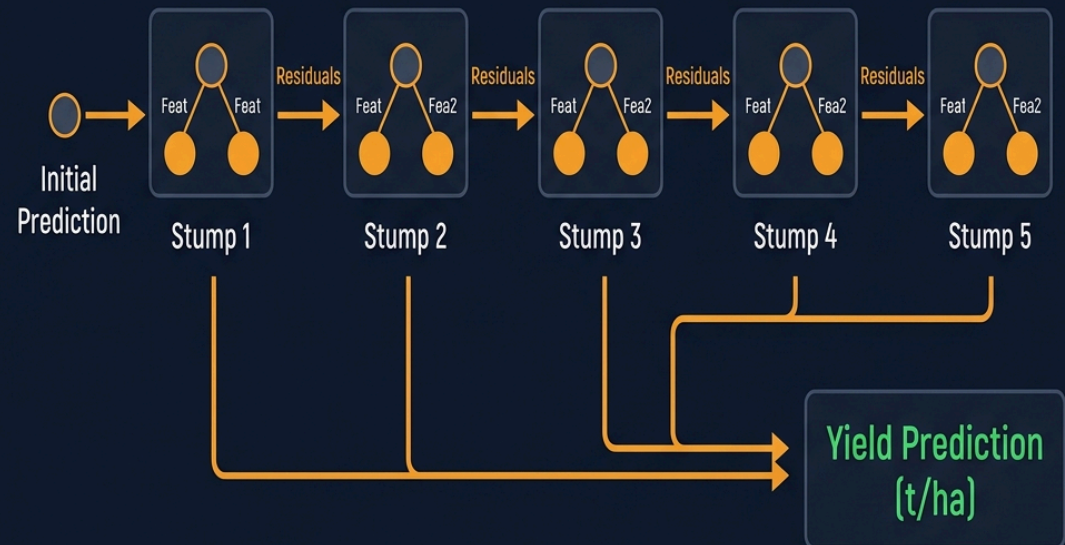
Algorithm Steps

- Initialize with mean of all target yields
- For each of 50 estimators:
 - a) Compute residuals (actual – prediction)
 - b) Fit a decision stump (depth 4) on residuals
 - c) Predictions += learning_rate × stump output
- Confidence = $1 - (\text{RMSE} / \text{target std deviation})$
- Output: yield (t/ha), confidence, RMSE

Key Params

Parameter	Value
Estimators	50
Learning Rate	0.1
Max Depth	4
Min Samples/Split	5
Loss	MSE

Gradient Boosting sequential Learning



Fertilizer Types & Nutrient Content

7 types with NPK composition and real market cost data

Fertilizer	N (%)	P (%)	K (%)	Cost (₹/kg)	Best For
Urea	46	0	0	8	High N deficiency
DAP	18	46	0	28	N + P deficiency
NPK 10-26-26	10	26	26	22	P + K deficiency
NPK 20-20-20	20	20	20	25	Balanced needs
MOP	0	0	60	18	High K deficiency
SSP	0	16	0	10	Phosphorus deficiency
Amm. Sulphate	21	0	0	12	Acidic soils

Fertilizer Selection Logic

Rule-based training label assignment — decision tree flowchart



Optimization Layer

Calculating optimal quantity, cost, and environmental impact score

Quantity Optimization

- Compute N,P,K deficits vs crop profile
- $Qty = \text{max deficit} / \text{nutrient\%} \times 100$
- Capped at 20–500 kg/hectare
- Scales by user's land area

Cost Estimation

- $\text{Cost} = \text{quantity} \times \text{cost-per-kg}$
- Per-hectare and total cost
- 7 fertilizers with real market rates
- Supports ₹ INR currency

Environmental Impact

- Over-application penalty (+15)
- High rainfall leaching risk (+20)
- Soil acidification risk (+15)
- Moisture runoff risk (+10)
- Qty-proportional impact (+0–40)

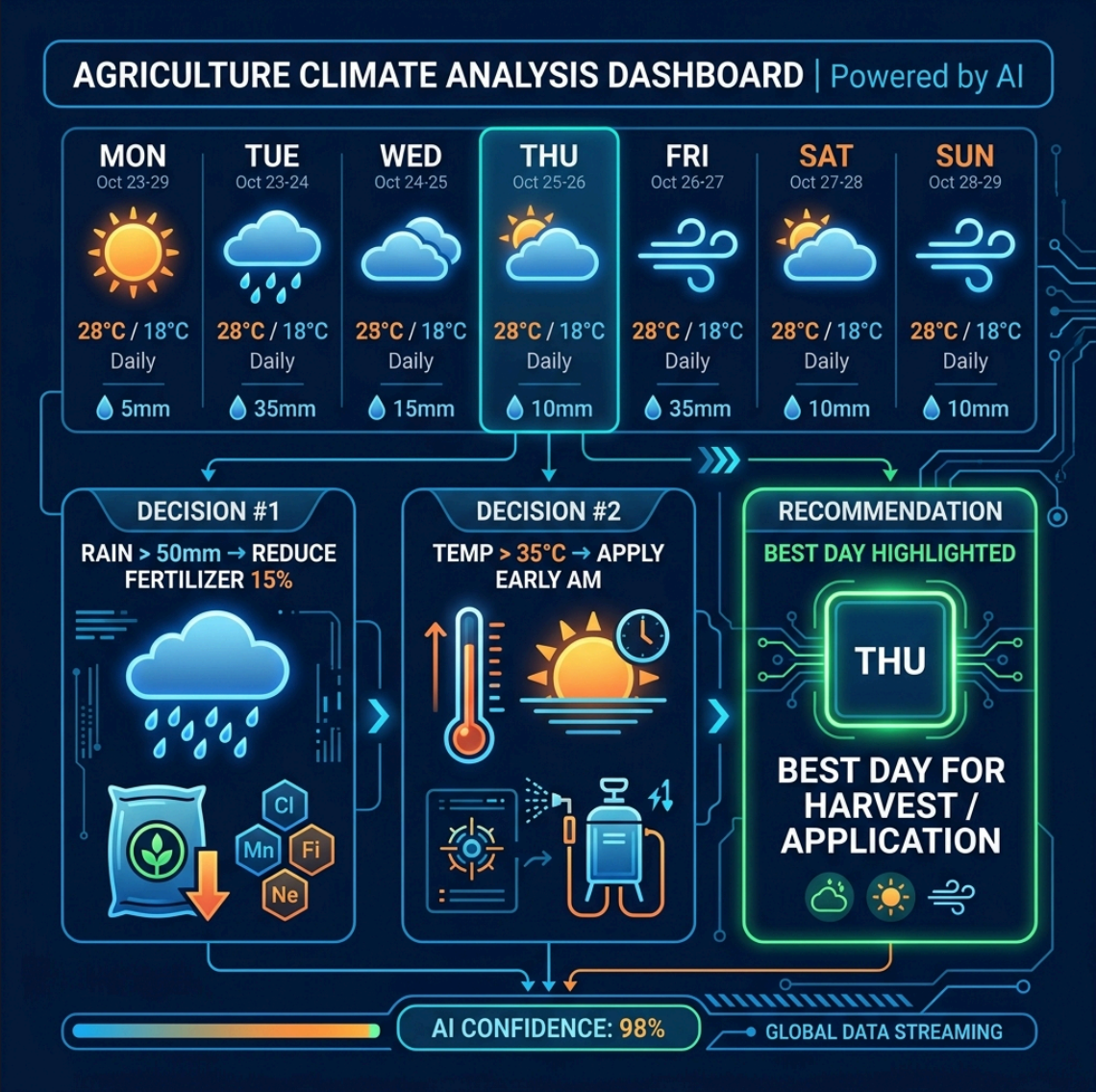
Climate-Aware Decision Engine

7-day weather forecast adjusts quantity and application timing

☁️ Climate Logic

- Generates 7-day forecast: temp, rainfall, humidity, wind
- Analyzes next 5 days for best application window
- Heavy rain (>50mm) → reduce quantity 15%, delay
- Moderate rain (>25mm) → reduce 8%, split doses
- High temp (>35°C) → apply early morning only
- High humidity (>80%) → monitor fungal disease risk
- Best Application Day — lowest risk window is scored

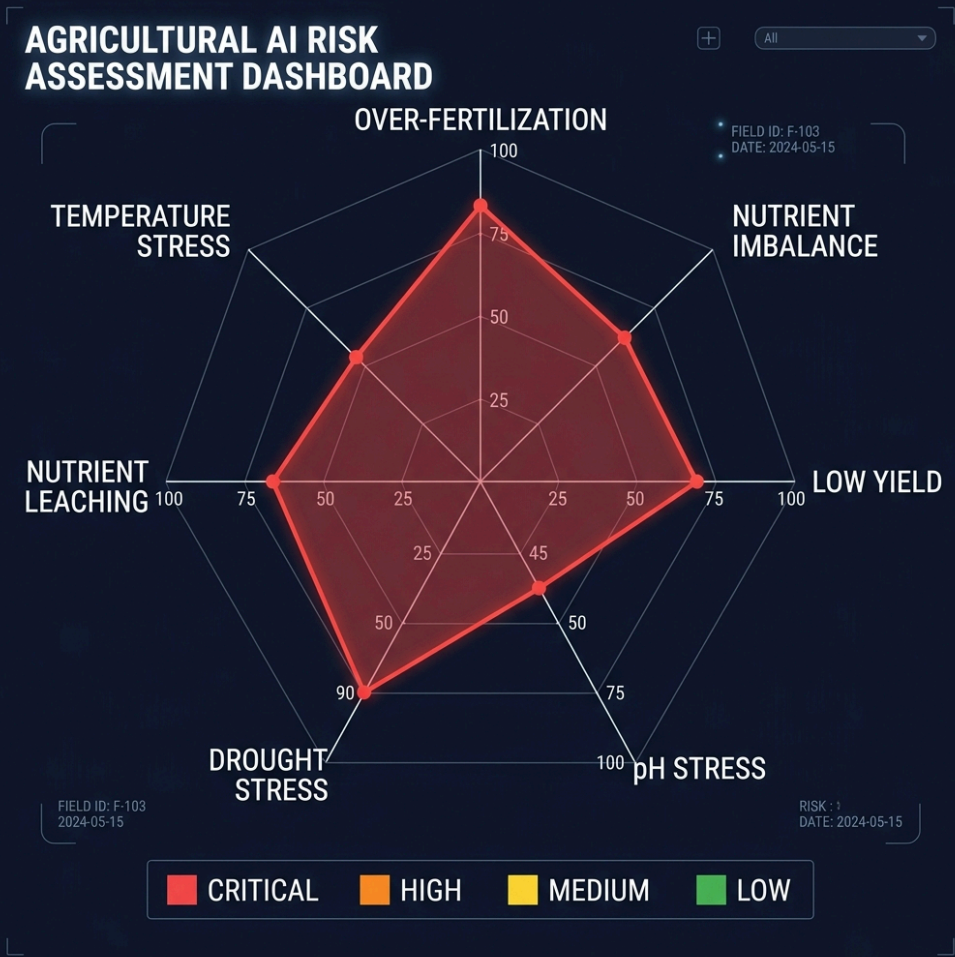
Condition	Qty Adjust	Risk
Rain >50mm	-15%	High
Rain 25-50mm	-8%	Medium
Temp >35°C	-10%	Medium
Humidity >80%	No change	Low
Dry + Moderate	Normal	Low



Risk Prediction Module

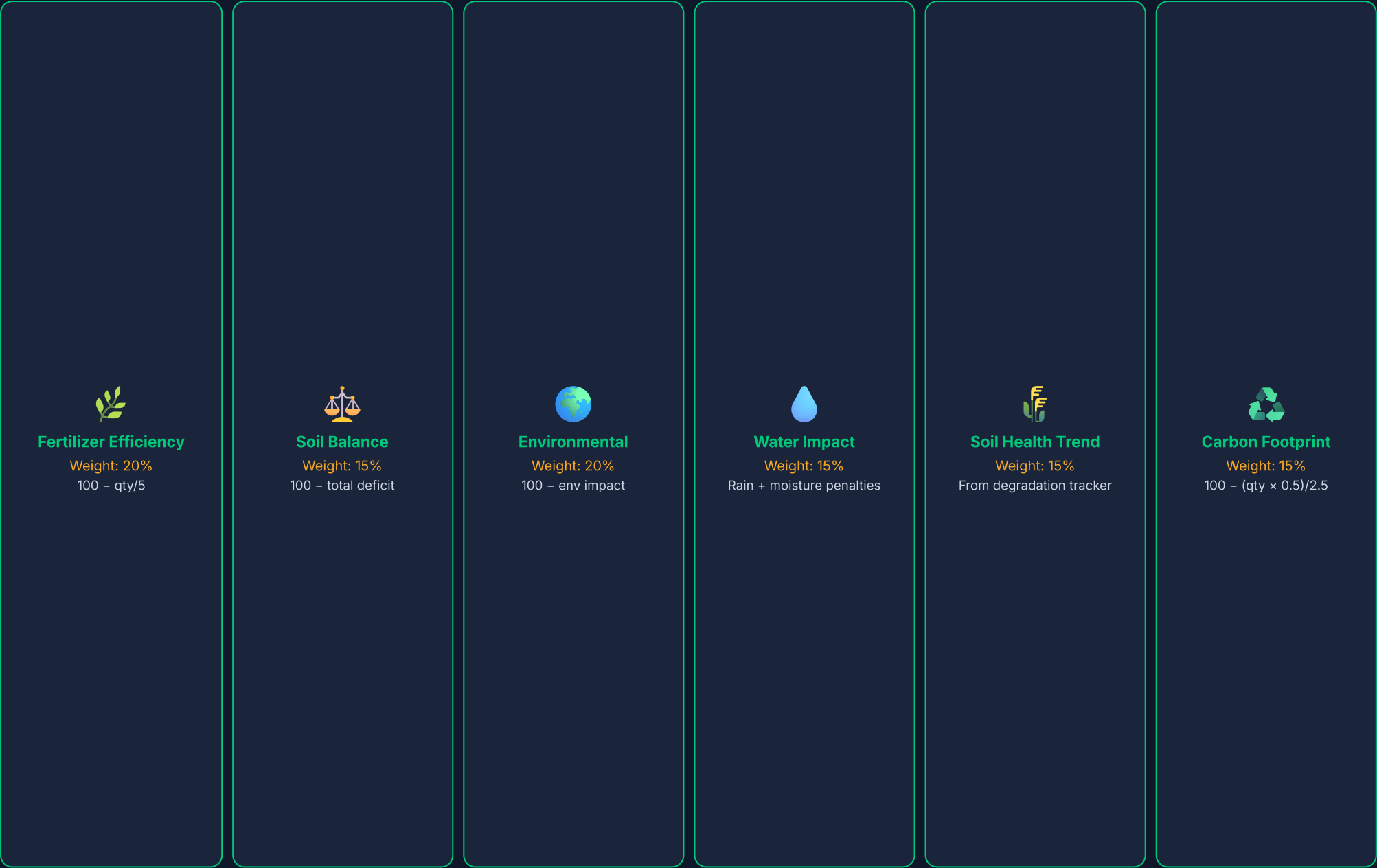
7 risk categories with severity, probability, and actionable recommendations

Risk Category	Trigger Condition	Severity	Recommendation
⚠️ Over-Fertilization	NPK excess >20 kg/ha	High/Med	Reduce, soil test
⚖️ Nutrient Imbalance	N:P:K ratio gap >0.6	High/Med	Apply balanced NPK
📉 Low Yield	Predicted <60% of avg	High/Med	Review soil + irrigation
🌱 pH Stress	pH outside optimal ±1	High/Med	Lime or Sulfur
🌧️ Drought Stress	Moisture <25%, Rain <30	High/Med	Irrigate + mulch
💧 Leaching	Rain >200mm, Qty >100	High/Med	Slow-release doses
🌡️ Temp Stress	Temp outside ±5°C	Medium	Shade / row covers



Sustainability Index

6-component weighted scoring system with grades A–F



Grades: A (≥80) Excellent | B (≥65) Good | C (≥50) Fair | D (≥35) Poor | F (<35) Critical

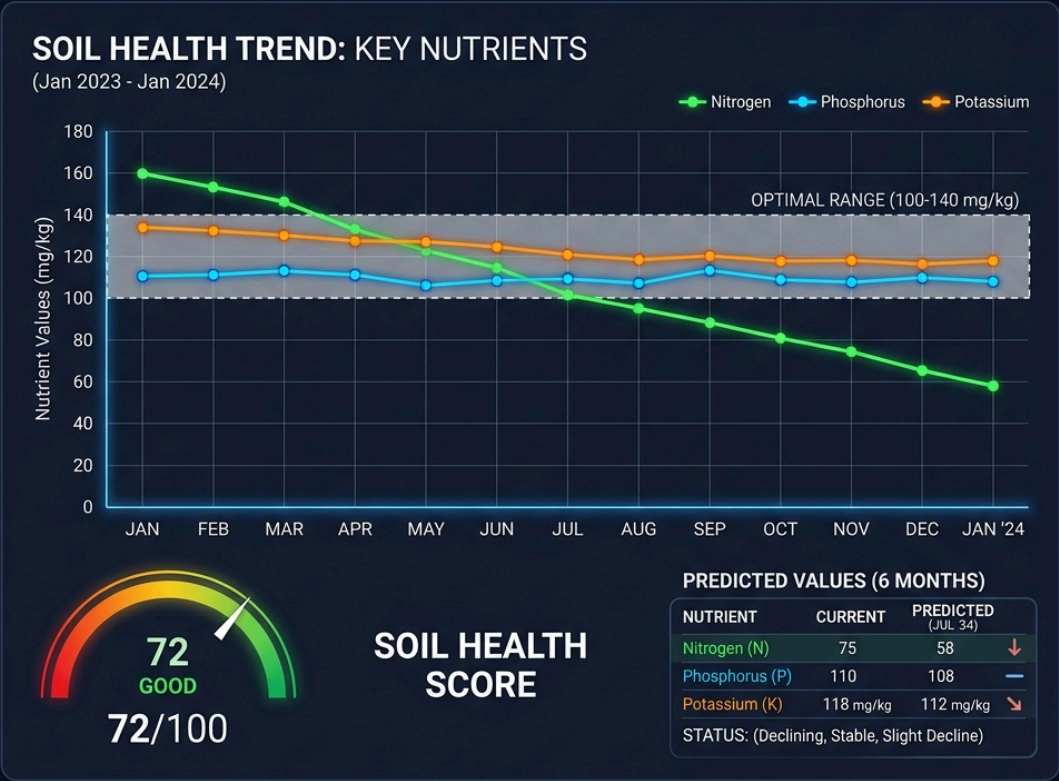
Soil Health Degradation Tracker

Linear trend projection — predicts nutrient decline over 6+ months

How It Works

- Records soil N, P, K, pH, moisture for each analysis
- Builds time-series history (up to 100 entries) in localStorage
- Linear regression on nutrient values over time index
- Projects values 6+ months into the future
- Classifies trend: Declining / Stable / Improving
- Health score (0–100) from nutrient proximity to optimal
- Generates urgency-based recommendations

Nutrient	Optimal Low	Optimal High
Nitrogen	40 kg/ha	120 kg/ha
Phosphorus	30 kg/ha	70 kg/ha
Potassium	25 kg/ha	60 kg/ha
pH	5.5	7.5



Adaptive Learning & Feedback Loop

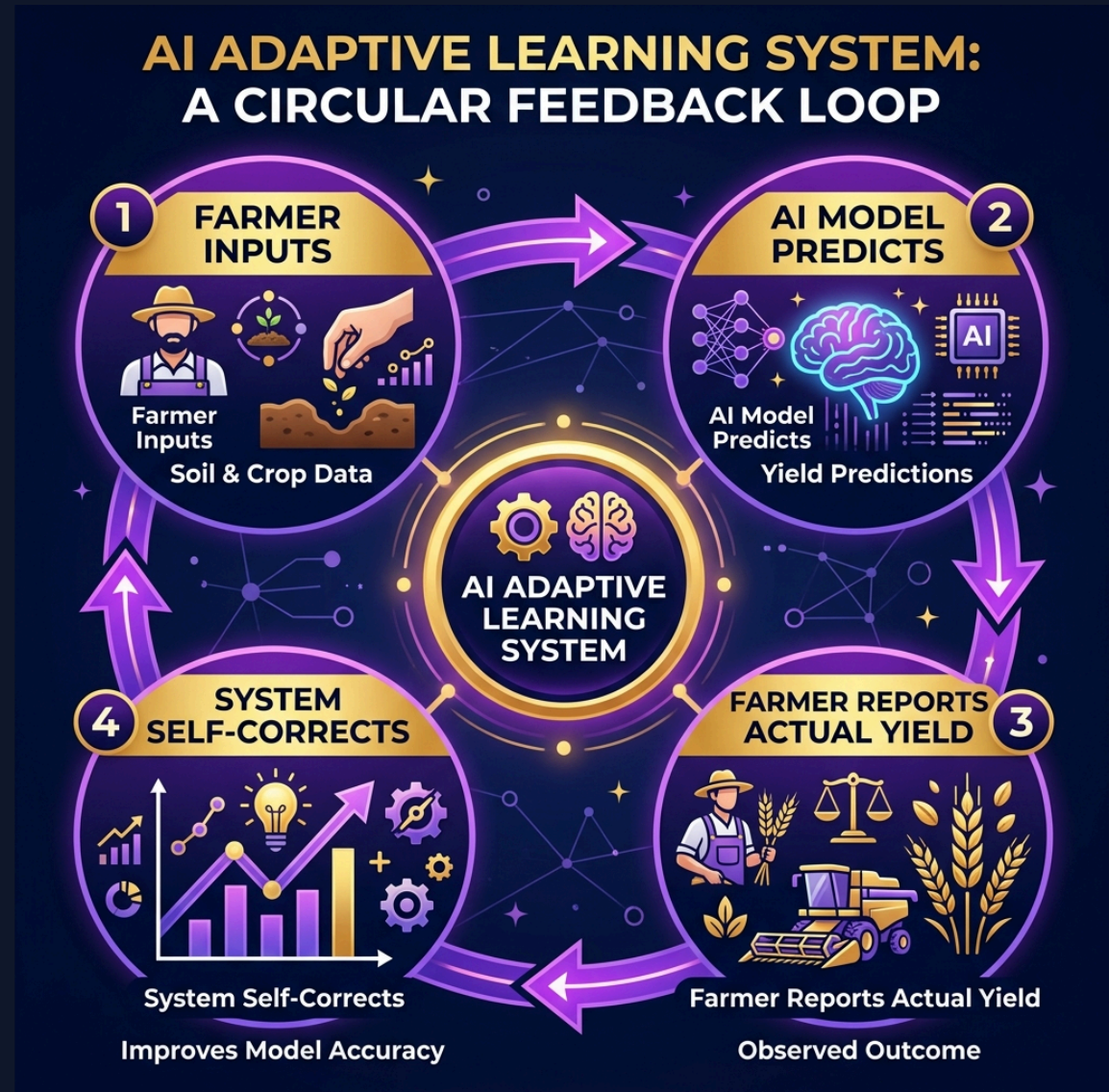
Continuous self-improvement through farmer feedback

Adaptive Learning

- Records every recommendation (input + predicted output)
- Stores up to 200 recommendations in localStorage
- Correction factor = $\text{actual_yield} / \text{predicted_yield}$ (per crop)
- Blends: 70% model + 30% learned correction
- Blend weight grows with more feedback (max 30%)
- Accuracy tracked: within 20% error = 'correct'
- Builds personal crop profiles over time

Farmer Feedback

- Submit: actual yield, fertilizer used, satisfaction (1-5)
- Status: Awaiting → Learning (1+) → Active (5+ feedbacks)
- Trend: first-half vs second-half accuracy tracking



Scenario Simulator — What-If Analysis

Explore impact of different conditions on yield, cost, and sustainability

Scenario	Modifications Applied	Use Case
 Reduce Fertilizer 30%	N, P, K × 0.7	Cost saving analysis
 Increase Fertilizer 30%	N, P, K × 1.3	Yield maximization
 Drought Conditions	Rainfall=20mm, Moisture=20%	Drought preparedness
 Heavy Rainfall	Rainfall=350mm, Moisture=85%	Flood risk planning
 Heat Wave	Temp=42°C, Humidity=30%	Extreme heat response
 Optimal Conditions	All params set to ideal values	Best-case benchmarking

Each scenario computes: Δ Yield (t/ha) | Δ Cost (₹) | Δ Environmental Impact | Natural-language recommendation

Explainability & Smart Alerts

Every recommendation comes with human-readable reasoning and risk alerts

💡 Model Explainability

- Feature importance via correlation analysis
- Top-3 influencing features shown per recommendation
- Nutrient-specific explanations (N, P, K deficit context)
- Environmental context notes
- Yield context: above/below average for that crop type
- Sustainability badges earned

🔔 Smart Alerts

Level	Trigger	Message
 Critical	pH < 4.5	Apply lime immediately
 Critical	Moisture < 20%	Irrigate before applying
 Warning	pH < 5.5	Acidic soil detected
 Warning	pH > 8.5	Apply gypsum to correct
 Warning	N < 20 kg/ha	Yellowing leaves risk
 Warning	Temp > 40°C	Apply in early morning
 Info	Low deficits	Minimal fertilizer needed

Technology Stack & Key Metrics

Engineering details and performance characteristics

⚡ Technology Stack

- Frontend: Vanilla JavaScript (ES6 Modules)
- Styling: CSS3 with Glassmorphism design
- Build: Vite bundler and dev server
- ML: Custom Random Forest + Gradient Boosting (no libraries)
- Storage: localStorage for adaptive learning + history
- Charts: Custom SVG / Canvas charting
- Deployment: Static hosting (Vercel / Netlify)
- Zero server dependencies — 100% in browser

📊 Key Metrics

Metric	Value
Training Samples	2,000 synthetic
Feature Dimensions	18 (8 numeric + 10 encoded)
Supported Crops	10
Fertilizer Classes	7
RF Trees	15
GB Estimators	50
Risk Categories	7
Sustainability Components	6
Code Modules	12

Future Roadmap

Planned enhancements for the next development phases

Phase 1

Real Weather API

Connect to IMD / OpenWeather for live 7-day forecast data

Phase 2

Satellite Imagery

NDVI from Sentinel-2 for real-time crop health monitoring

Phase 3

Backend + Auth

Node.js / Firebase for cloud data persistence and user accounts

Phase 4

Mobile App

Offline-capable React Native app for rural farmers with SMS alerts

Thank You!

AgriML — Empowering Farmers with AI

Questions & Discussion

 Smart Fertilizer •  Yield Prediction •  Climate Aware •  Sustainable